

CHAPTER 1

STRUCTURE AND FUNCTION OF BODY SYSTEMS

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After completing this chapter, you will be able to

- describe both the macrostructure and microstructure of muscle and bone;
- describe the sliding-filament theory of muscular contraction;
- describe the specific morphological and physiological characteristics of different muscle fiber types and predict their relative involvement in different sport events;
- describe the anatomical and physiological characteristics of the cardiovascular and respiratory systems; and
- predict the cardiorespiratory responses to different types of exercise or sports events.

Physical exercise and sport performance involve effective, purposeful movements of the body. These movements result from the forces developed in muscles, which move the various body parts by acting through lever systems of the skeleton. Skeletal muscle activation

initiates in upper motor neurons located in the cerebral cortex, and activation signals are relayed to lower motor neurons residing in the spinal cord that are part of the peripheral nervous system. Support for this neuromuscular activity involves continuous delivery of oxygen and nutrients to working tissues and removal of carbon dioxide and metabolic waste by-products from working tissues through activities of the cardiovascular and respiratory systems.

In order to best apply the available scientific knowledge to the training of athletes and the development of effective training programs, strength and conditioning professionals must have a basic understanding of not only musculoskeletal function but also those systems of the body that directly support the work of exercising muscle. Accordingly, this chapter summarizes those aspects of the anatomy and function of the musculoskeletal, neuromuscular, cardiovascular, and respiratory systems that are essential for developing and maintaining muscular force and power.

Musculoskeletal System

The musculoskeletal system of the human body consists of bones, joints, muscles, and tendons configured to allow the great variety of movements characteristic of human activity. This section describes the various components of the musculoskeletal system, both individually and in the context of how they function together.

Skeleton

The muscles of the body function by pulling against bones that rotate about joints. Muscles can only pull, not push, given that the activation of muscles results in shortening contractions. However, through the system of bony levers, muscle contractions can be manifested as either pulling or pushing forces against external objects.

The body has approximately 206 bones, though the number can vary. This relatively light, strong structure provides leverage, support, and protection to internal organs ([figure 1.1](#)). The **axial skeleton** consists of the skull (cranium), vertebral column (vertebra C1 through the coccyx), ribs, and sternum. The **appendicular skeleton** includes the shoulder (or pectoral) girdle (left and right scapula and clavicle); bones of the arms, wrists, and hands (left and right humerus, radius, ulna, carpals, metacarpals, and phalanges); the pelvic girdle (left and right coxal or innominate bones); and the bones of the legs, ankles, and feet (left and right femur, patella, tibia, fibula, tarsals, metatarsals, and phalanges).

Junctions of bones are called *joints*. **Fibrous joints** (e.g., sutures of the skull) allow virtually no movement, **cartilaginous joints** (e.g., intervertebral disks) allow limited movement, and **synovial joints** (e.g., elbow and knee) allow considerable movement. Movements in relation to sports and exercise mainly occur at synovial joints, whose most important features are low friction and large range of motion. Articulating bone ends are covered with smooth **hyaline cartilage**, and the entire joint is enclosed in a

capsule filled with **synovial fluid**. Additional supporting structures of ligament and cartilage usually exist (41).

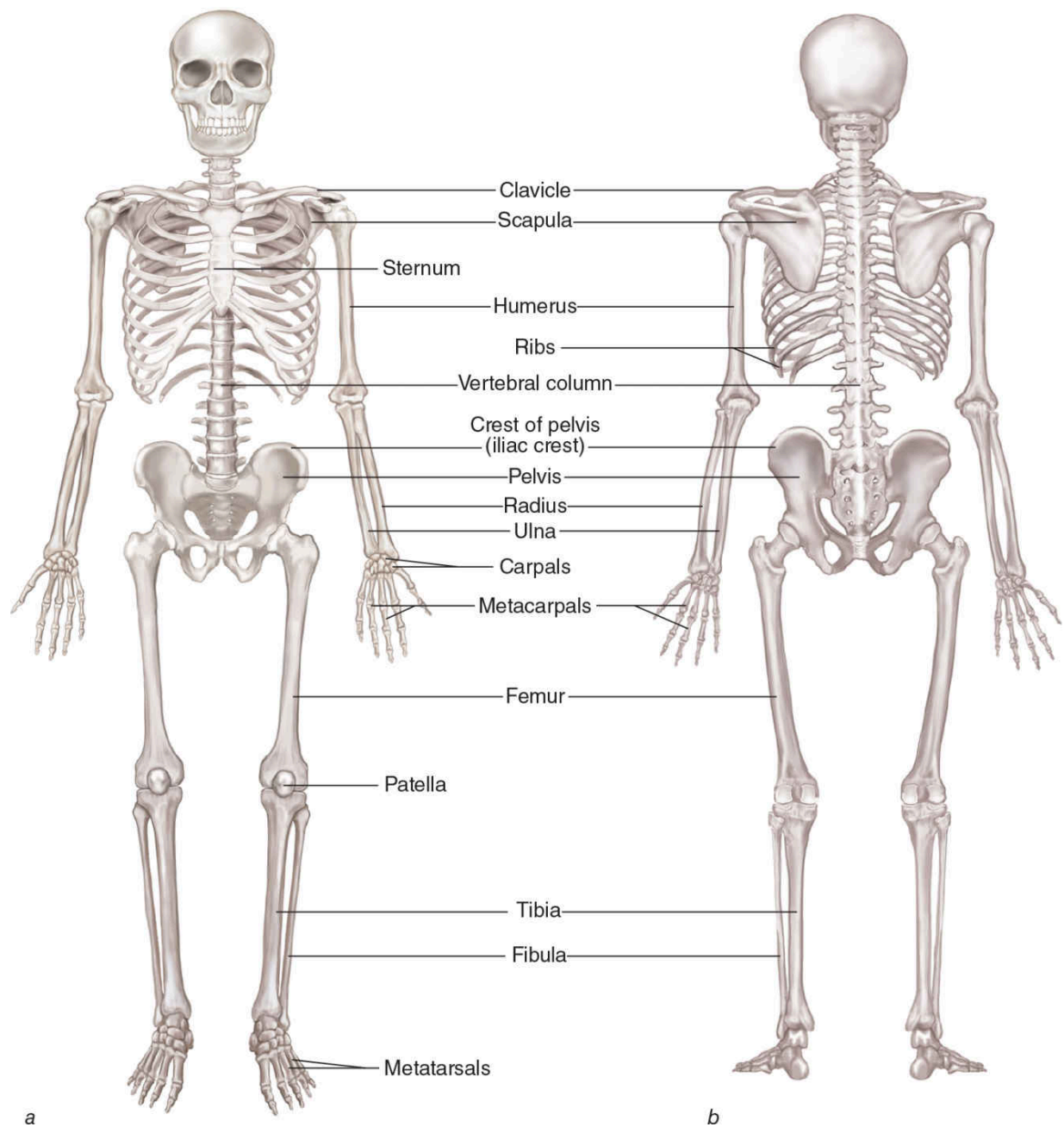


FIGURE 1.1 (a) Front view and (b) rear view of an adult male human skeleton.

What Factors Affect Skeletal Growth in an Adult?

Aside from genetics, dietary habits and physical activity patterns affect the adult skeleton. Several research studies have supported resistance training as yielding positive bone adaptations. Hence, consistently lifting heavy loads during job tasks or via resistance training increases bone density and bone mineral content. Consistently performing more explosive movements with impact facilitates similar adaptations. Some of the highest bone densities have been observed in people who engage in gymnastics or other activities that involve high-strength, high-power, or high-impact movements (29). Since bone adaptations occur less rapidly compared to skeletal muscle, the consistent implementation of progressive overload is important to facilitate positive adaptations.

Virtually all joint movements consist of a rotation about points or axes. Joints can be categorized by the number of directions about which rotation can occur. **Uniaxial joints**, such as the elbow, operate as hinges, essentially rotating about only one axis. The knee is often referred to as a *hinge joint*, but its axis of rotation actually changes throughout the joint range of motion. **Biaxial joints**, such as the ankle and wrist, allow movement about two perpendicular axes. **Multiaxial joints**, including the shoulder and hip ball-and-socket joints, allow movement about all three perpendicular axes that define space.

The **vertebral column** is made up of vertebral bones separated by flexible disks that allow movement to occur. The vertebrae are grouped into 7 cervical vertebrae in the neck region; 12 thoracic vertebrae in the middle to upper back; 5 lumbar vertebrae in the lower back; 5 sacral vertebrae, which are fused together and make up the rear part of the pelvis; and 3 to 5 coccygeal vertebrae, which form a kind of vestigial internal tail extending downward from the pelvis.

Skeletal Musculature

The system of muscles that enables the skeleton to move is depicted in [figure 1.2](#). The connection point between bones is the joint, and skeletal muscles are attached to bones at each of their ends. Without this arrangement, movement could not occur.

Musculoskeletal Macrostructure and Microstructure

Each skeletal muscle contains muscle tissue made up of a variety of cell types (with the most notable being muscle fibers), connective tissue, nerves, blood vessels, and a plethora of cell types between muscle fibers that serve different functions. Fibrous connective tissue, or **epimysium**, covers the body's more than 430 skeletal muscles. The epimysium is contiguous with the tendons at the ends of the muscle ([figure 1.3](#)). The **tendon** is attached to **bone periosteum**, a specialized connective tissue covering all bones. Notably, muscle contractions pull on tendons, and in turn, this moves bones that are attached to the associated tendons. Limb muscles have two attachments to bone, including **proximal** (closer to the trunk) and **distal** (farther from the trunk) attachments. The two attachments of trunk muscles are termed **superior** (closer to the head) and **inferior** (closer to the feet).

Muscle cells, often called **muscle fibers** or **myofibers**, are long (sometimes running the entire length of a muscle), cylindrical cells 50 to 100 μm in diameter (about the diameter of a human hair). Muscle fibers have all their nuclei situated on the outer periphery of the cell and have a striated appearance when viewed with a microscope under low magnification. Under the epimysium the muscle fibers are grouped in bundles (**fasciculi**) that may consist of up to 150 fibers, with the bundles surrounded by connective tissue called **perimysium**. Each muscle fiber is surrounded by connective tissue called **endomysium**, which is encircled by and is contiguous with the muscle fiber membrane, or **sarcolemma** (41). All the connective tissue—epimysium, perimysium, and endomysium—is contiguous with the tendon, so tension developed in a muscle cell is

transmitted to the tendon and the bone to which it is attached ([figure 1.3](#)).

The junction between a **motor neuron** (nerve cell) and the muscle fibers it innervates is called the *motor end plate*, or, more often, the **neuromuscular junction** ([figure 1.4](#)). Each muscle cell has only one neuromuscular junction, although a single motor neuron innervates many muscle fibers, sometimes hundreds or even thousands. A motor neuron and the muscle fibers it innervates are called a **motor unit**. All the muscle fibers of a motor unit contract together when they are stimulated by the motor neuron.

The interior structure of a muscle fiber is depicted in [figure 1.5](#). The **sarcoplasm**, which is the cytoplasm of a muscle fiber, is an aqueous medium through which the exchange of ions (sodium, potassium, calcium) and metabolites (ADP, ATP, inorganic phosphate) occur to and from different organelles. Not only is the sarcoplasm of critical importance to muscle fibers because of this role, but intracellular organelles and structures are also suspended in the sarcoplasm. These organelles and structures include the contractile apparatus constructed of thick and thin protein filaments, other proteins including metabolic enzymes and signaling proteins, stored glycogen and fat particles, ribosomes that facilitate protein synthesis, and specialized organelles such as mitochondria and the sarcoplasmic reticulum.

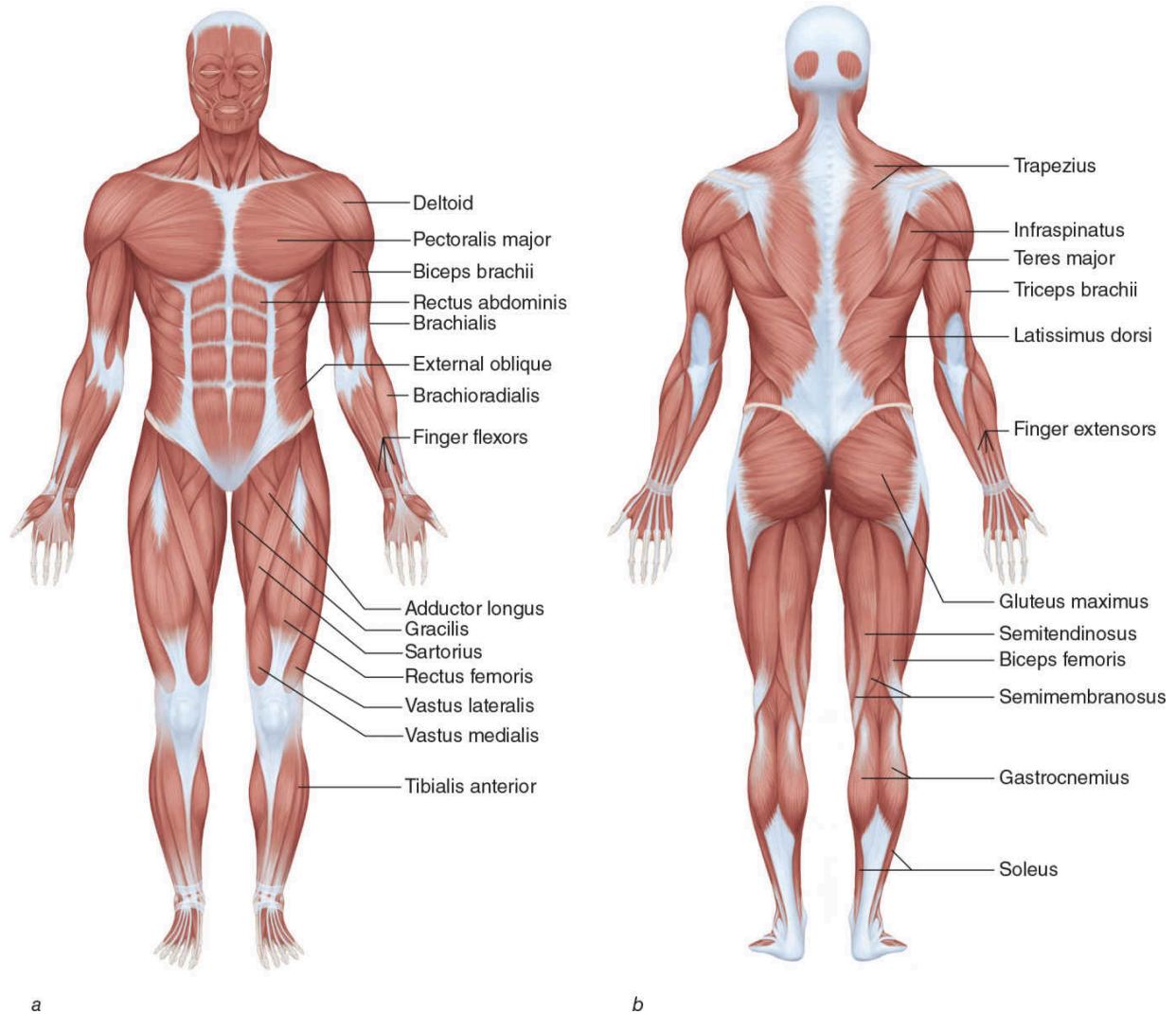


FIGURE 1.2 (a) Front view and (b) rear view of adult male human skeletal musculature.

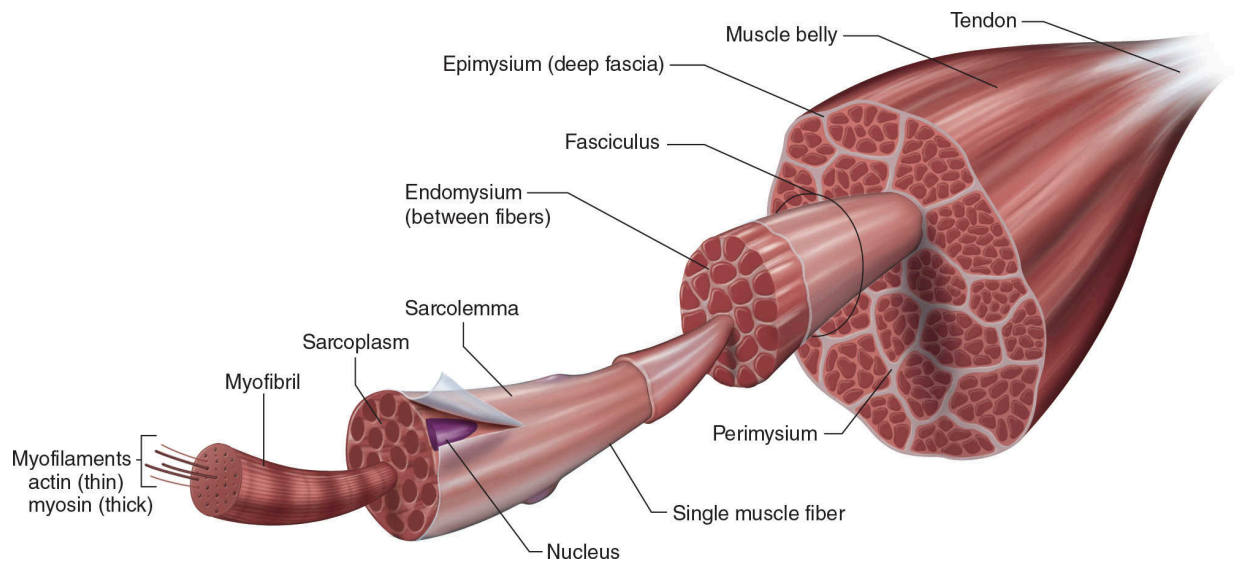


FIGURE 1.3 Schematic drawing of a muscle illustrating three types of connective tissue: epimysium (the outer layer), perimysium (surrounding each fasciculus, or group of muscle fibers), and endomysium (surrounding individual muscle fibers).

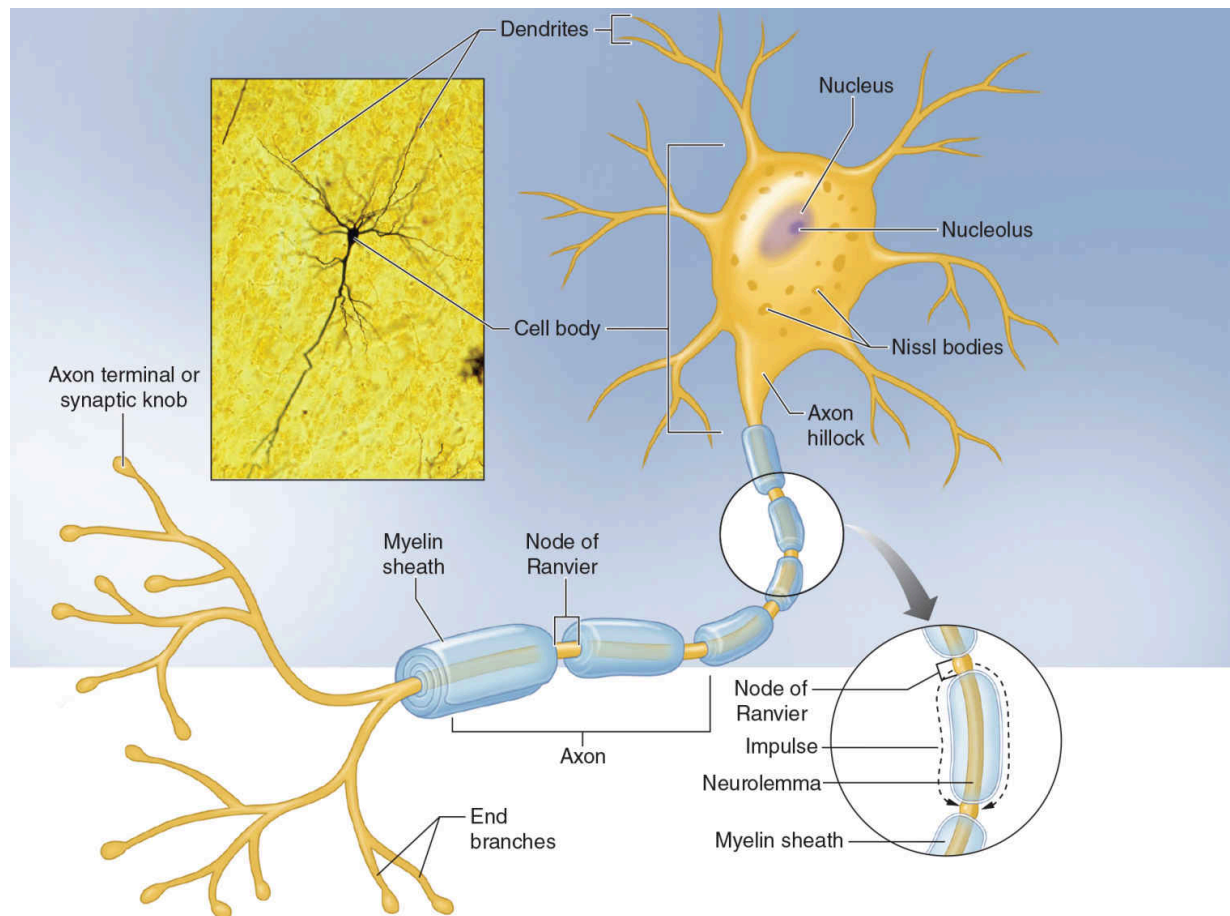


FIGURE 1.4 Motor unit, consisting of a motor neuron and the muscle fibers it innervates. A single motor unit typically contains several hundred muscle fibers.

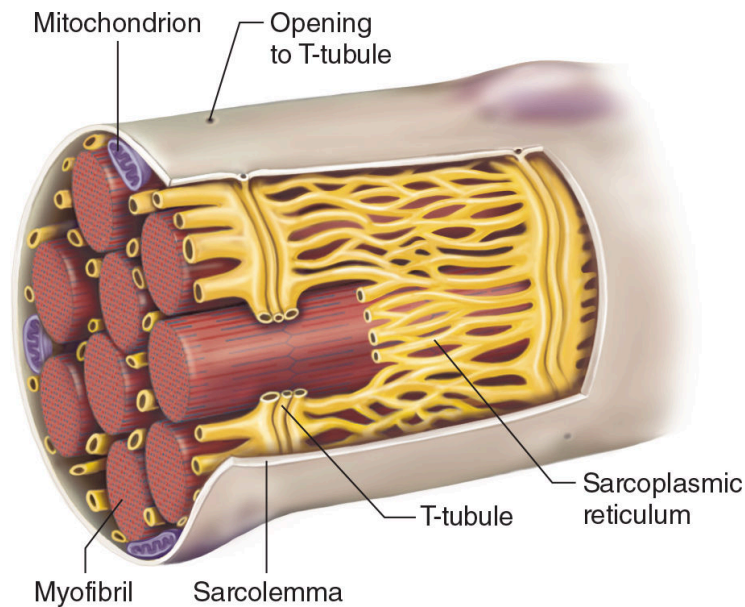


FIGURE 1.5 Sectional view of a muscle fiber.

Hundreds of **myofibrils** (each about $1\text{ }\mu\text{m}$ in diameter, $1/100$ the diameter of a hair) exist in the sarcoplasm of one muscle fiber. Myofibrils are the contractile machinery of muscle fibers, and two types of **myofilaments** are highly enriched in myofibrils, including **myosin** and **actin**. The thicker myosin filaments (about 16 nm in diameter, or $1/10,000$ the diameter of a hair) contain up to 200 myosin molecules. A single myosin molecule consists of two myosin heavy chain proteins and four myosin light chain proteins. This molecule also contains two head-like features (one being from each of the two heavy chains), two hinge points below each head, and two tail-like features that act to keep the two heavy chain proteins bound. When examining a thick filament containing hundreds of myosin molecules, the globular heads protrude away from the thick filament at regular intervals, and the head-like structures form **crossbridges** with actin filaments during muscle contraction. The thin actin filaments (about 6 nm in diameter) consist of two strands arranged in a double helix. Myosin and actin filaments are organized longitudinally in the smallest contractile unit of skeletal muscle termed the **sarcomere**. Sarcomeres average about $2.5\text{ }\mu\text{m}$ in length in a relaxed fiber (approximately 4,500 per centimeter of muscle

length) and are repeated through the entire length of myofibrils that span the muscle fiber (4).

The arrangement of the myosin and actin filaments within sarcomere boundaries give muscle fibers a striated dark/light appearance under magnification. The dark **A-band** corresponds with the alignment of the myosin filaments, whereas the light **I-band** corresponds with the areas in two adjacent sarcomeres that contain only actin filaments (41). The **Z-line** is in the middle of the I-band and appears as a thin, dark line running longitudinally through the I-band. The **H-zone** is the area in the center of the sarcomere, where only myosin filaments are present. During muscle contraction, the H-zone decreases as the actin filaments slide over myosin filaments toward the center of the sarcomere. The I-band also decreases as the Z-lines are pulled toward the center of the sarcomere.

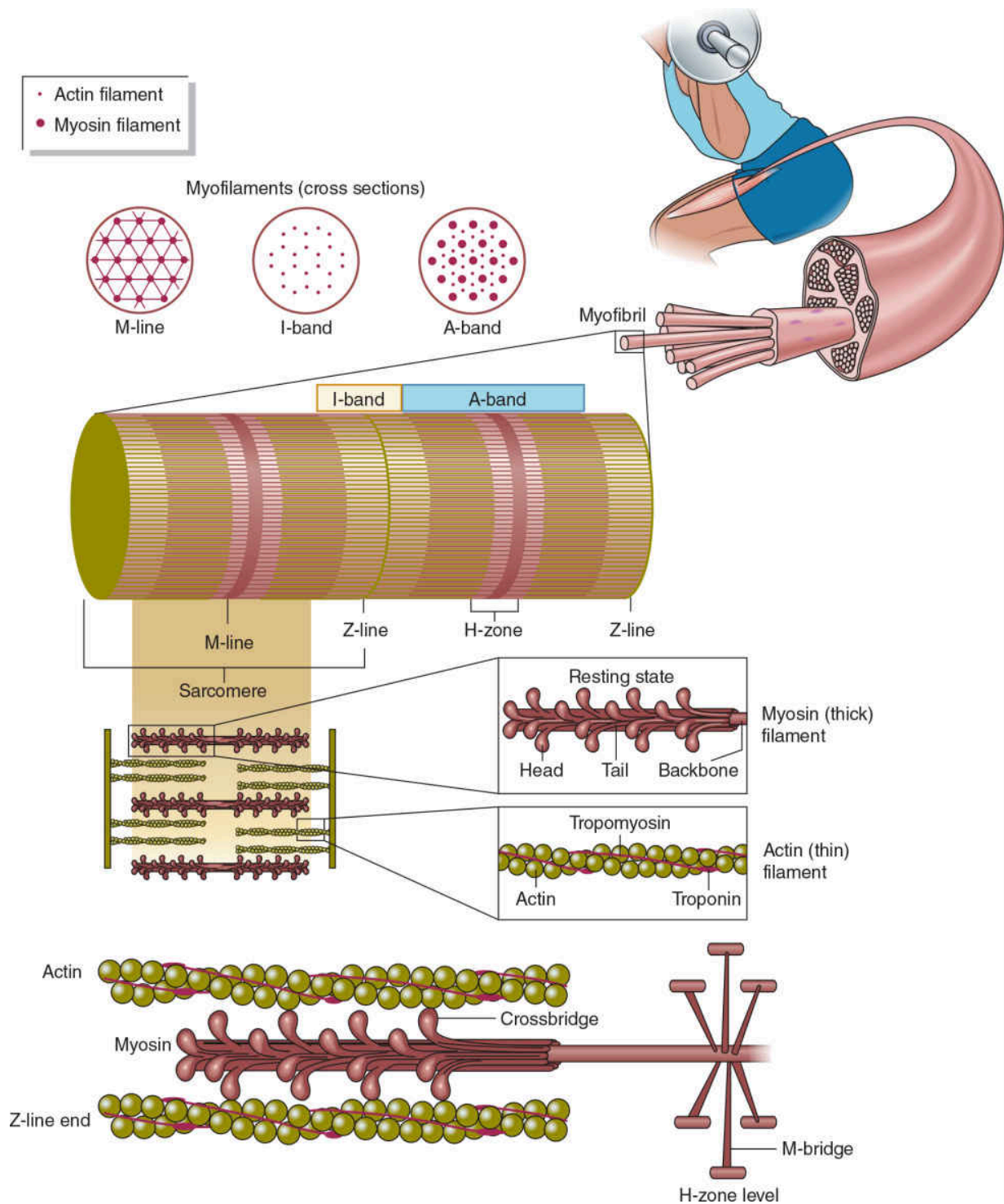


FIGURE 1.6 General view of the myosin and actin filaments in muscle. The arrangement of myosin (thick) and actin (thin) filaments gives skeletal muscle its striated appearance.

[Figure 1.6](#) shows a simplistic depiction of myosin and actin in the sarcomere. Biochemical bonds allow for myosin proteins to aggregate into the highly organized thick myosin filament, and thick filaments are anchored to specialized proteins in the M-line, which is in the center of the sarcomere (the center of the H-zone). Actin filaments are anchored to Z-lines through another set of specialized proteins. Although both M-lines and Z-lines exist in tandem throughout the entire myofibril, individual sarcomeres are defined as the thick myosin and thin actin filaments as well as associated proteins found between two Z-lines. It is also notable that six actin filaments surround each myosin filament, and each actin filament is surrounded by three myosin filaments. This lattice-like arrangement enables a highly organized and efficient structure for muscular contraction.

Each myofibril is encased by an intricate system of tubules, called the **sarcoplasmic reticulum** ([figure 1.5](#), [page 5](#)), which terminates as vesicles in the vicinity of the Z-lines. Calcium ions are stored in the sarcoplasmic reticulum, and muscle activation leads to the release of these calcium stores to facilitate muscular contraction. Discharge of an **action potential** (an electrical nerve impulse) arrives nearly simultaneously from the surface to all depths of the muscle fiber. Calcium is thus released throughout the muscle fiber, producing a coordinated contraction of the entire cell.

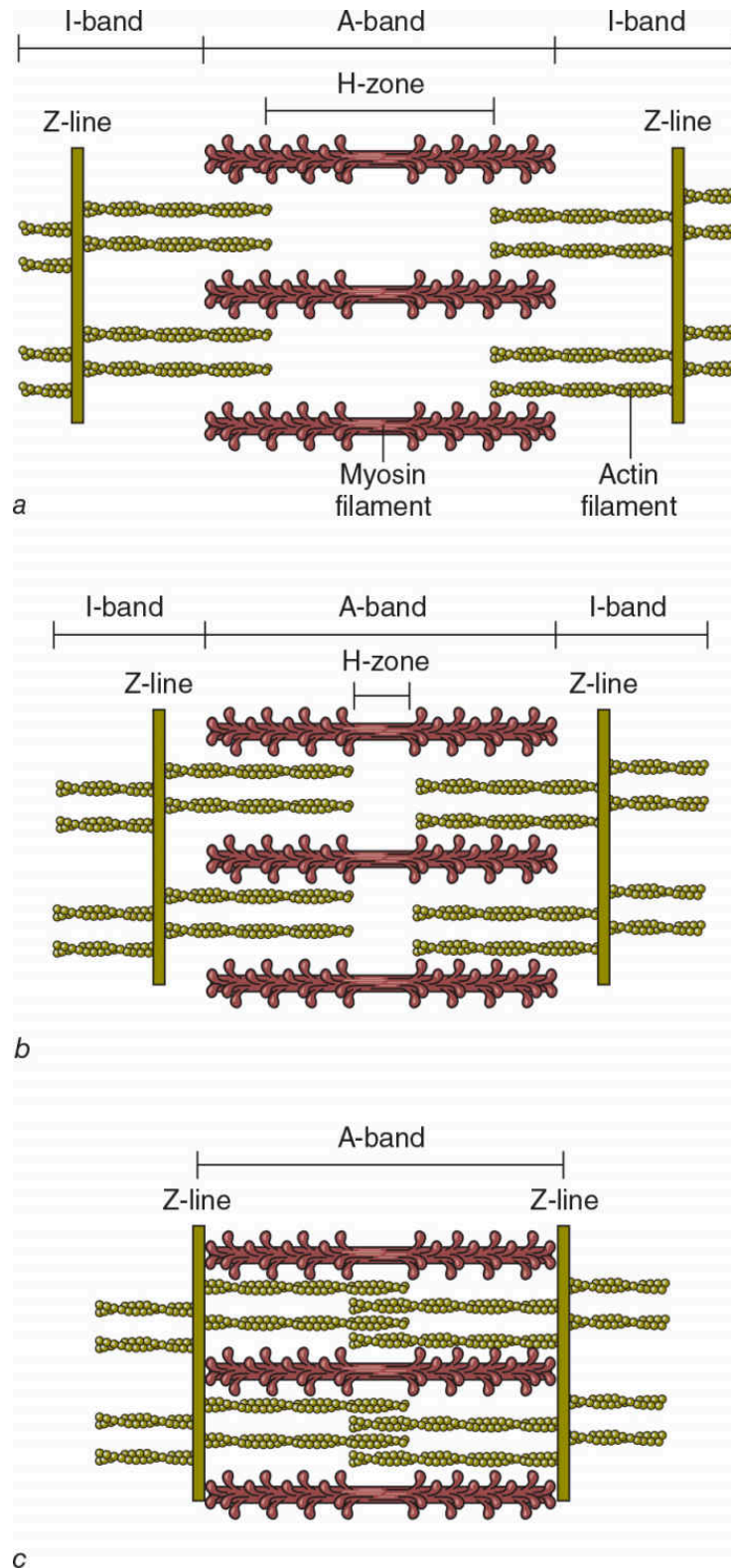


FIGURE 1.7 Contraction at the sarcomere level. (a) In stretched muscle the I-bands and H-zone are elongated and there is low force potential due to a reduction in actin and myosin overlap and, thus, actin-myosin cross-bridge formation potential. (b) When muscle

contracts (here partially), the I-bands and H-zone are shortened. In this example, force potential is highest due to optimal actin and myosin overlap. (c) With contracted muscle, force potential is again low because actin and myosin overlap is not optimal.

The discharge of an action potential from a motor neuron signals the release of calcium from the sarcoplasmic reticulum into the myofibrils of a muscle fiber, leading to a cascade of events that result in the contraction of the muscle fiber.

Sliding-Filament Theory of Muscular Contraction

The **sliding-filament theory** was proposed by Huxley and Hanson in 1954 to mechanistically explain muscular contraction at the molecular level (22). In its simplest form, the sliding-filament theory indicates that actin filaments anchored to each end of the sarcomere slide inward on myosin filaments. This results in Z-lines being pulled toward the center of the sarcomere and in muscle fiber shortening ([figure 1.7](#)). As actin filaments slide over myosin filaments, both the H-zone and I-band shrink. The energy-dependent association of myosin heads with actin filaments (i.e., actin-myosin crossbridge formation), as well as the energy-dependent inward movement of actin filaments over the myosin filaments through what is termed the **power stroke**, are the bases for muscle contraction. Although a very small displacement of an actin filament occurs with each power stroke, this action is rapidly repeated across all sarcomeres during voluntary muscle contractions leading to appreciable movement with prolonged contractions (41).

Resting Phase

When a muscle is not active, low levels of calcium are present in the myofibrils of muscle fibers (most of it is stored in the sarcoplasmic reticulum), so very few actin-myosin crossbridges are formed.

Excitation–Contraction Coupling Phase

When a motor neuron activates a muscle fiber, a cascade of events occur that stimulate the sarcoplasmic reticulum to release calcium ions into the myofibrils of the muscle fiber. Secreted calcium ions bind with a high affinity to the **troponin** protein, which exists at regular intervals along the actin filament ([figure 1.6](#)). This calcium-troponin interaction causes a shift in another protein termed **tropomyosin**, which is bound to troponin and spans the length of the actin filament. These events expose the myosin binding site on actin and enable the myosin head to attach to the actin filament to form an actin-myosin crossbridge (4). It is important to understand that the amount of force produced by any given muscle is directly related to the number of actin-myosin crossbridges formed within the muscle fibers of that muscle (4). Thus, activities that require a high force output necessitate the activation (or recruitment) of most motor neurons and the muscle fibers they innervate to achieve the maximal amount of actin-myosin crossbridge formation.

The number of actin-myosin crossbridges formed within the muscle fibers of a given muscle at any instant in time dictates the force production of the muscle.

Contraction Phase

The contraction phase involves the mechanical movement (or pivoting) of the myosin head from the resting phase when the actin-myosin crossbridge is formed, which leads to the pulling of the associated actin filament to the center of the sarcomere. Critically, this mechanical movement is catalyzed from the biochemical energy provided by the hydrolysis (or breakdown) of adenosine triphosphate (ATP) by the **myosin adenosine triphosphatase** (ATPase) enzyme present on the myosin head. The products of ATP hydrolysis include adenosine diphosphate (ADP) and inorganic phosphate (P_i), metabolites that also serve critical roles in the contraction process. While the different phases of the contraction phase are relatively complex, they can be summarized as follows:

1. The myosin head hydrolyzes ATP into ADP and P_i , both of which remain bound to the myosin head. The energy provided by this biochemical reaction results in a conformational change whereby the myosin head is cocked into a high-energy state.
2. In its high-energy state, the myosin head binds to actin to form the actin-myosin crossbridge.
3. The release of P_i from the myosin head catalyzes a conformational change in the myosin molecule, leading to the power stroke, which is the pivoting of the myosin head that pulls the actin filament toward the center of the sarcomere.
4. Following the first power stroke, ADP is released from the myosin head and another ATP molecule binds to the myosin head.
5. The actin-myosin crossbridge is subsequently dissociated, the myosin head returns to its original low-energy state and conformation, and the process is repeated with the hydrolysis of the newly bound ATP molecule.

So long as calcium ions remain bound to troponin during muscle activation, contiguous rapid rounds of actin-myosin crossbridge formation, power strokes, and actin-myosin crossbridge dissociation will occur, and this is termed **crossbridge cycling**.

Calcium and ATP are necessary for crossbridge cycling with actin and myosin filaments.

Relaxation Phase

Relaxation occurs when the stimulation of the motor neuron ceases. Calcium is pumped back into the sarcoplasmic reticulum through specialized transporters that also require ATP, and the restoration of low calcium ion concentrations in myofibrils highly reduces the number of actin-myosin crossbridges.

Neuromuscular System

Muscle fibers are innervated by motor neurons that transmit impulses in the form of electrochemical signals from the spinal cord to the muscle. A motor neuron generally has numerous terminal branches at the end of its axon, allowing for several muscle fibers to be innervated. This integrated structure is termed the *motor unit*. Motor units can be generally described as slow-twitch, fast-twitch/fatigue-resistant, and fast-twitch/fatigable. These motor unit characteristics dictate muscle fiber type, contractile characteristics, and involvement in different types of activities and exercise.

Activation of Muscles

When a motor neuron fires an impulse or action potential, all the muscle fibers that it innervates are simultaneously activated and contract through the mechanisms previously described. The number of muscle fibers vary within each motor unit. Muscles that are required to function with great precision, such as eye muscles, may have motor units with as few as one muscle fiber. Changes in the number of active motor units in these small muscles can produce the extremely fine gradations in force that are necessary for precise movements of the eyeball. In contrast, the quadriceps muscle group, which moves the leg with much less precision, may have several hundred muscle fibers innervated by one motor neuron.

An action potential is an electrochemical signal that travels along a motor neuron from the cell body to the nerve terminal. However, this signal is not capable of directly exciting muscle fibers. Instead, the arrival of the action potential at nerve terminals of the motor neuron causes the secretion of a neurotransmitter called **acetylcholine** to each of the neuromuscular junctions contained within the motor unit. Secreted acetylcholine binds to and activates acetylcholine receptors present on the sarcolemma of muscle fibers within the motor unit. Once enough acetylcholine receptors are activated, action potentials are generated along the sarcolemma of

these muscle fibers. These action potentials continue down T-tubules to eventually catalyze the release of calcium from the sarcoplasmic reticulum as discussed earlier in this chapter. These events ultimately lead to all the muscle fibers within the motor unit to simultaneously contract and develop force. There is no evidence to date suggesting that an active motor unit only activates a portion of the muscle fibers to contract. Similarly, no evidence to date suggests that individual muscle fibers can be partially activated within an active motor unit. Hence, the **all-or-none principle** has been generally used to describe this phenomenon; specifically, the activation of any given motor neuron will lead to the activation and contraction of all of the muscle fibers it innervates.

Each action potential traveling down a motor neuron results in the rapid activation of the muscle fibers within the motor unit. The brief contraction that occurs in response to one action potential is referred to as a **twitch**. Although calcium release from the sarcoplasmic reticulum during a twitch is sufficient to allow for actin-myosin crossbridging to occur, a brief twitch (which lasts milliseconds) does not enable a high level of force to be developed given that calcium is rapidly sequestered back to the sarcoplasmic reticulum and the muscle fiber relaxes ([figure 1.8a](#)). If a second twitch is elicited from the motor neuron before the innervated muscle fibers completely relax, forces generated from the two twitches summate, resulting in a greater force relative to that which is produced by a single twitch ([figure 1.8b](#)). Decreasing the time interval between twitches results in an even greater force summation, and a high activation frequency leads to the merging of twitches in what has been termed **unfused** and **fused tetanic contractions** ([figure 1.8c and d](#)). The force generated by a motor unit during fused tetanic contractions (i.e., the maximal tetanic force) is the maximal force that the motor unit can develop (8). The ability of motor units to sustain this type of contraction (along with the size of muscle fibers and motor unit recruitment patterns) is likely one of the key determinants of maximal force and power outputs during athletic endeavors.

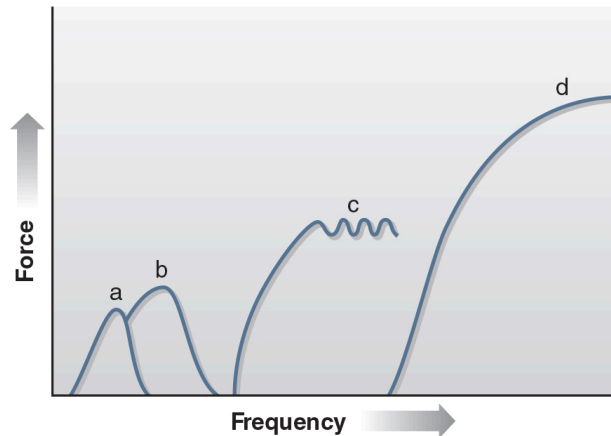


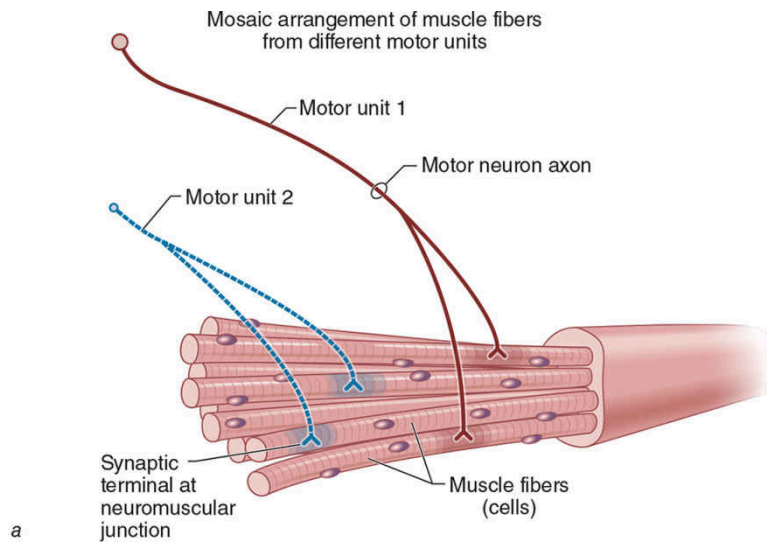
FIGURE 1.8 Twitch, twitch summation, and tetanus of a motor unit: *a* = single twitch; *b* = force resulting from summation of two twitches; *c* = unfused tetanus; *d* = fused tetanus.

Muscle Fiber Types

Skeletal muscles are composed of muscle fibers that have markedly different morphological and physiological characteristics (37). These differences have led to different classification systems, with the most familiar including the terms **slow-twitch** and **fast-twitch** muscle fibers. Because a motor unit innervates muscle fibers that are metabolically and functionally similar to one another, motor units have also been historically designated using this nomenclature. A fast-twitch motor unit is one that develops force and relaxes rapidly, thus exhibiting a shorter twitch time. Slow-twitch motor units, in contrast, develop force and relax slowly and have a longer twitch time. General differences between these types of motor units can be found in [figure 1.9](#).

Regarding muscle fiber classification, histochemical staining for myosin ATPase content was also used for years to classify myofibers as **type I** (slow-twitch), **type IIa** (fast-twitch), and **type IIx** (fast-twitch). Other methods used for myofiber classification involve single fiber isolation techniques to quantify the presence of different myosin heavy chain isoforms and immunohistochemical staining for myosin heavy chain isoforms (33). The nomenclature for both techniques is akin to that associated with the myosin ATPase histology methods.

Fiber types are also delineated by differences in fatigue resistance and force production capabilities, and this can be due to small to moderate differences in cellular morphology, intracellular protein content, and organelle distribution. For instance, type I fibers are generally more fatigue resistant relative to type II fibers, and type I fibers have a higher capacity to produce ATP via aerobic metabolism. This difference can be largely attributed to mitochondrial characteristic differences between fiber types, because type I fibers have a higher density and a more even distribution of mitochondria between myofibrils relative to type II fibers (31). However, type I fibers produce less overall force and yield a lower rate of force development compared to type II fibers. These latter characteristic differences are chiefly due to type I myofibers exhibiting lower cellular diameters (and thus less myofibrils per cell) compared to type II fibers (21). Additionally, myosin ATPase reaction rates differ between the type I and II isoforms, wherein the type I isoform exhibits a relatively lower reaction rate and thus a reduced rate at which crossbridge cycling can occur (38).



	Slow-twitch motor unit	Fast-twitch motor unit
Conduction velocity	slow	fast
Fiber type	type I	type IIa or IIx
Number of muscle fibers innervated	lower	higher
Recruitment threshold	lower	higher
Fatigue resistance	higher	lower

b

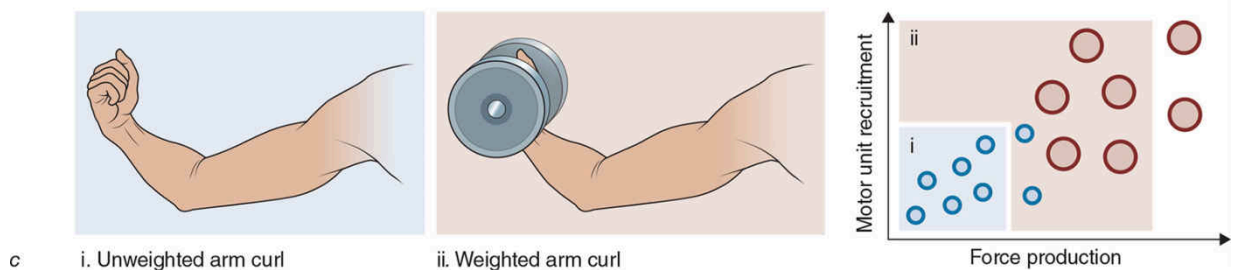


FIGURE 1.9 Comparison of slow- and fast-twitch motor units: *a* = schematic of slow-twitch and fast-twitch motor units innervating separate sets of muscle fibers; *b* = characteristics of slow-twitch and fast-twitch motor units; *c* = hypothetical differences between unweighted versus weighted arm curls on slow-twitch and fast-twitch motor unit recruitment patterns. *Note:* In healthy individuals this innervation pattern (*a*) exists in a pseudo-random mosaic pattern. Unlike non-weighted contractions that recruit lower threshold motor units, weighted contractions (*c*) will recruit both lower threshold slow-

twitch and higher threshold fast-twitch motor units, which is an example of the size principle of motor unit recruitment.

Other notable differences exist between type I and II fibers. For instance, type I fibers have more contacts with capillaries compared to type II fibers (5, 19, 23, 34), and this feature (along with smaller cellular diameters and more intracellular myoglobin) yields a higher oxygen diffusion potential that favors aerobic metabolism. A 2021 research study using sophisticated molecular approaches has also unveiled that the content of hundreds of structural and metabolic proteins differs between type I, IIa, and IIx fibers (30), and these findings suggest that these appreciable differences contribute to the unique fiber type characteristics discussed in this paragraph. Based on these differences, it is not surprising that postural muscles, such as the soleus, have a high composition of type I fibers, whereas larger locomotor muscles, such as the quadriceps muscles, have a mixture of both type I and type II fibers to enable both low and high power output activities (such as jogging and sprinting, respectively). Refer to [table 1.1](#) for a summary of the primary characteristics of fiber types.

Motor units are composed of muscle fibers with specific morphological and physiological characteristics that determine their functional capacity.

TABLE 1.1 Major Characteristics of Muscle Fiber Types

Characteristic	Fiber types		
	Type I	Type IIa	Type IIx
Motor neuron size	Small	Large	Large
Recruitment threshold	Low	Intermediate/high	High
Nerve conduction velocity	Slow	Fast	Fast

Characteristic	Fiber types		
	Slow	Fast	Fast
Contraction speed	Slow	Fast	Fast
Relaxation speed	Slow	Fast	Fast
Fatigue resistance	High	Intermediate/low	Low
Endurance	High	Intermediate/low	Low
Force production	Low	Intermediate	High
Power output	Low	Intermediate/high	High
Aerobic enzyme content	High	Intermediate/low	Low
Anaerobic enzyme content	Low	High	High
Sarcoplasmic reticulum complexity	Low	Intermediate/high	High
Capillary density	High	Intermediate	Low
Myoglobin content	High	Low	Low
Mitochondrial size, density	High	Intermediate	Low
Fiber diameter	Small	Intermediate	Large
Color	Red	White/red	White

Motor Unit Recruitment Patterns

Through everyday experiences, it is evident that a given muscle can vary its level of force output according to the level required by a particular task. This ability to vary or gradate force is essential for performance of smooth, coordinated patterns of movement. Muscular force can be graded in two ways. One is through variation in the frequency at which motor units are activated. A single twitch results from one action potential, and this does not produce a great deal of force. However, if the frequency of activation is increased so that the forces of the twitches begin to overlap or summate, the resulting force developed by the motor unit is much greater. This method of varying force output is especially important in small muscles, such as those of the hand. Even at low forces, most of the motor units in these muscles are activated, albeit at a low frequency.

Force output of the whole muscle is intensified through an increase in the frequency of firing of individual motor units. Another manner to increase muscle force involves increasing the number of motor units activated, a process known as *recruitment*. In large muscles, such as those in the thigh, most motor units are required to be activated at relatively higher frequencies during near-maximal contraction efforts. As a result, increases in force output are achieved through increases in motor unit firing frequency as well as the recruitment of additional motor units.

TABLE 1.2 Relative Involvement of Muscle Fiber Types in Sport Events

Event	Type I	Type II
100 m sprint	Low	High
800 m run	High	High
Marathon	High	Low
Olympic weightlifting	Low	High
Soccer, lacrosse, hockey	High	High
American football wide receiver	Low	High
American football lineman	Low	High
Basketball, team handball	Low	High
Volleyball	Low	High
Baseball or softball pitcher	Low	High
Mixed martial arts	High	High
Wrestling	High	High
50 m swim	Low	High
Field events	Low	High
Cross-country skiing, biathlon	High	Low
Tennis	High	High

Event	Type I	Type II
Downhill or slalom skiing	High	High
Speed skating	High	High
Bobsled	Low	High
Distance cycling	High	Low
Rowing	High	High

The types of motor units recruited for a given activity are determined by the physiological demand imposed by that activity over a given unit of time ([table 1.2](#)). For an activity such as slower long-distance running, slow-twitch motor units are predominantly engaged in working muscles given that physiological demand is not relatively high over time and these motor units possess a heightened endurance capacity and resistance to fatigue. Sprints at the end of a race that impose a relatively higher physiological demand, however, require the additional activation of fast-twitch motor units. If an activity requires near-maximal contractions over very short time periods, such as those observed with the power clean, most of the motor units in the involved muscle(s) are recruited and the fast-twitch motor units make more of a significant contribution to the effort. Electromyography estimates have indicated that maximal voluntary contractions with higher intensity weightlifting typically lead to approximately 90% or more activation of available motor units in the involved muscle(s) (7). However, some caveats must be considered with this estimation. First, the activation of the entire motor neuron pool is probably not possible in untrained people (9, 10, 18). Moreover, although large fast-twitch units may be recruited if the effort is substantial, it is probably not possible to activate them at a high enough frequency under most circumstances for maximal force to be realized. These details aside, it is important to appreciate that motor unit activation varies regarding different types of physical activity.

The force output of a muscle can be varied through changes in the frequency of activation of individual motor units as well as a change in the number of activated motor units.

Proprioception

Proprioceptors are specialized sensory receptors located within joints, muscles, and tendons. Because these receptors are sensitive to pressure and tension, they relay information concerning muscle dynamics to the conscious and subconscious parts of the central nervous system. The brain is thus provided with information concerning kinesthetic sense, or conscious appreciation of the position of body parts with respect to gravity. Most of this proprioceptive information, however, is processed at subconscious levels so we do not have to dedicate conscious activity toward tasks such as maintaining posture or position of body parts.

Proprioceptors are specialized sensory receptors that provide the central nervous system with information needed to maintain muscle tone and perform complex coordinated movements.

Muscle Spindles

Muscle spindles are proprioceptors that consist of several modified muscle fibers enclosed in a sheath of connective tissue ([figure 1.10](#)). These modified fibers, called **intrafusal fibers**, run parallel to the normal, or **extrafusal**, fibers. Muscle spindles provide information concerning muscle length and the rate of change in length. When the muscle lengthens, spindles are stretched. This deformation activates the sensory neuron of the spindle, which sends an impulse to the spinal cord, where it synapses (connects) with motor neurons. This results in the activation of motor neurons that innervate the same muscle. Spindles thus indicate the degree to which the muscle

must be activated in order to overcome a given resistance. As a load increases, the muscle is stretched to a greater extent, and engagement of muscle spindles results in greater activation of the muscle. Muscles that perform precise movements have many spindles per unit of mass to help ensure exact control of their contractile activity. A simple example of muscle spindle activity is the knee jerk reflex. Tapping on the tendon of the knee extensor muscle group below the patella stretches the muscle spindle fibers. This causes activation of extrafusal muscle fibers in the same muscle. A knee jerk occurs as these fibers actively shorten. This, in turn, shortens the intrafusal fibers and causes their discharge to cease.

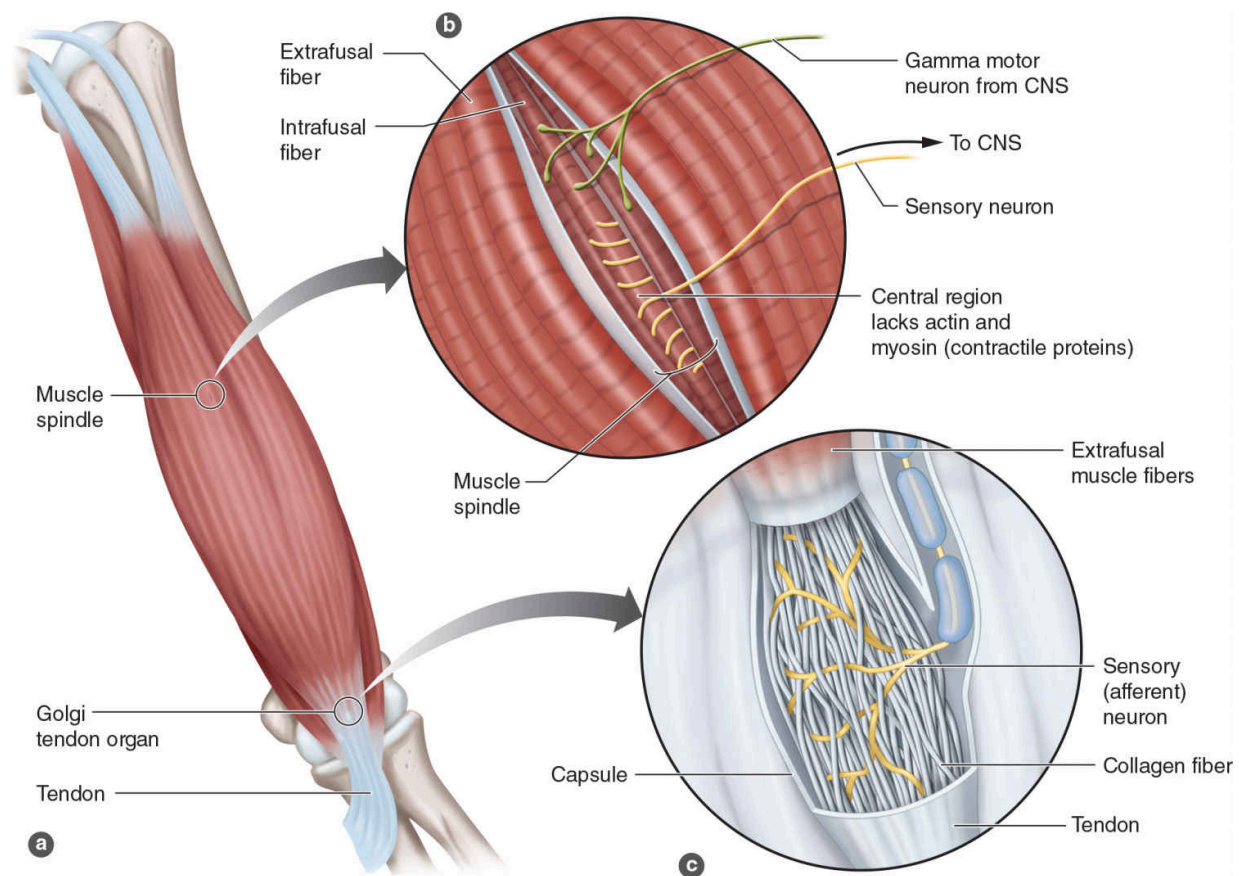


FIGURE 1.10 Muscle spindle. When a muscle is stretched, deformation of the muscle spindle activates the sensory neuron, which sends an impulse to the spinal cord, where it synapses with a motor neuron, causing the muscle to contract.

How Can Athletes Improve Force Production?

- Incorporate phases of training that use heavier loads (or explosive actions) to optimize motor unit recruitment and firing frequency patterns
- Also incorporate phases of training that promote skeletal muscular hypertrophy given that more myofibrils (and thus more actin-myosin crossbridges) will be available for near-maximal muscle contractions
- Finally, implement training methods that specifically target movements where increases in force are desired (e.g., heavy bench press and back squat exercises for increasing pushing forces)

Golgi Tendon Organs

Golgi tendon organs (GTOs) are proprioceptors located in tendons near the myotendinous junction and are in series—that is, attached end to end, with extrafusal muscle fibers ([figure 1.10](#)). Golgi tendon organs are activated when the tendon attached to an active muscle is stretched. As tension in the muscle and tendons increases, the discharge of the GTOs increases. The sensory neuron of the GTO synapses with an inhibitory interneuron in the spinal cord, which in turn synapses with and inhibits a motor neuron that serves the same muscle. The result is a reduction in tension within the muscle and tendon. Thus, whereas spindles facilitate activation of the muscle, neural input from GTOs inhibits muscle activation. The GTOs' inhibitory process is thought to provide a mechanism that protects against the development of excessive tension. The effect of GTOs is therefore minimal at low forces, but when an extremely heavy load is placed on the muscle, reflexive inhibition mediated by the GTOs causes the muscle to relax. The ability of the motor cortex to override this inhibition may be one of the fundamental adaptations to heavy resistance training.